Yijin Zeng

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Summary

Final year PhD candidate in statistics and machine learning at Imperial College London (StatML CDT). Research focuses on statistical hypothesis testing, statistical association, and change point detection in the presence of missing data. Strong statistics and machine learning foundations with hands on experience in end-to-end data science, machine learning, and deep learning projects; proficient in Python and R.

Education

- **Imperial College London**, London, UK — Oct. 2021– Oct. 2025 (Estimated)
  PhD in Statistics and Machine Learning (StatML CDT, Imperial & Oxford)
  - *Funded by Roth Scholarship*
- **Imperial College London**, London, UK — Oct. 2020 – Sep. 2021
  MSc in Statistics
  - *Degree classification: Distinction*
- **Beijing Institute of Technology**, Beijing, China — Sep. 2016 – Jun. 2020
  BSc in Mathematics
  - *Top 10% of cohort; Thesis with Distinction*

Work Experience

Tesco Data Scientist PhD Internship (London) — Aug.2024 – Oct.2024

- Worked with forecasting team and supply chain team, delivering an end-to-end machine learning solution for demand forecasting, including feature selections, data cleaning, model training, and evaluation.
- Designed and trained multiple deep neural network architectures for supply chain forecasting, achieving accuracy comparable or higher (5% $\sim$ 10% depending on the product categories) to Tesco's current approach with less workflows.

IQVIA Data Analyst Internship (Beijing) — Jul.2019 – Sep.2019

- Cleaned and analyzed survey and experiments datasets, performing feature engineering, missing-data handling, and statistical testing for supporting final experiments and survey conclusions.
- Worked on data visualization for efficiently communicating numerical results to managers and non-technical stakeholders.

Publications

- **Zeng Y**, Adams NM, Bodenham DA. On two-sample testing for data with arbitrarily missing values. arXiv preprint arXiv:2403.15327. 2024 Mar 22.
  - *A two-sample hypothesis testing method for partially observed univariate data to detect location shifts, with no assumption of the values of missing data.*
- **Zeng Y**, Adams NM, Bodenham DA. MMD Two-sample Testing in the Presence of Arbitrarily Missing Data. arXiv preprint arXiv:2405.15531. 2024 May 24.

- *Extended our framework for high-dimensional data with any distributional shifts.*

- **Zeng Y**, Adams NM, Bodenham DA. Exact Bounds of Spearman's footrule in the Presence of Missing Data with Applications to Independence Testing. arXiv preprint arXiv:2501.11696. 2025 Jan 20.
  - *Tight bounds for Spearman's footrule in the presence of missing data, with applications for independence testing.*

Open Source Packages

- **Zeng Y**, Bodenham D, and Adams N. Package 'wmwm'.
  - *R and Python packages for performing Wilcoxon-Mann-Whitney two sample hypothesis testing in the presence of missing data.*
- **Zeng Y**. Package 'bosfr'.
  - *R package for computing tight bounds of Spearman's footrule in the presence of missing data.*

Selected Data Science Projects

- E-Commerce Order Cancellation Prediction
  - *predict order cancellation with highly imbalanced dataset for real-world impact*
- Hangman Game
  - *play word guessing game using deep learning with high success rate (∼60%).*
- Stock Clustering
  - *cluster stocks using daily return for balanced results*

Selected Advanced Courses

- Machine Learning in Finance (by Dr. Mikko Pakkanen)
  - *machine learning algorithms for high-frequency trading (HFT)*
- Computational Optimal Transport and Statistical Learning with Missing Values (by Prof. Gabriel Peyré & Dr. Julie Josse)
  - *optimal transport for machine learning; dealing with missing data in machine learning*
- Selective Inference (by Dr. Daniel Garcia Rasines & Prof. Alastair Young)
  - *valid inference for 'post selection' problem in modern statistic and machine learning*

Selected Teaching Experience

- Probability for Statistics (MSc) — Summer term 2023
- Data Science (MSc) — Spring term 2024
- Statistical Clinics (MSc) — Summer term 2024
- Probability and Statistics (BSc) — Spring term 2024

Skills

- Language: Chinese (native), English (Proficient)
- Programming & Tools: Python, R, TensorFlow, pandas, NumPy, scikit-learn, MATLAB; Azure ML; HPC; Linux; Git.